



NUST

FACULTY OF COMPUTING AND INFORMATICS

DEPARTMENT OF SOFTWARE ENGINEERING

Private Bag 13388, 13 Storch Street, Windhoek, Namibia

Tel: +264-(0)61-207-2054 Fax: +264-(0)61-207-9054

Subject Code:	ASD810S
Subject:	Advanced Software Development

ASD810S Assignment

Required is a Postgraduate digital system for administering postgraduate studies at NUST. Task below form the bulk of the requirements:

- **Student** - uploads *Progress Report*, checks individual progress, feedback from Supervisor and the Higher Degrees Committee (HDC).
- **Supervisor** - submits *Summary of Proposals* (SoPs) or thesis on behalf of Student to the Head of Department (HoD). Also comment and sign on Student *Progress Report*. Grades their students' final thesis. Checks students' progress.
- **Internal Evaluators** - evaluates student's proposals before they sign the *Checklist*. Take note that this is the same as Supervisor but here has a different role.
- **External Evaluators** - grades students' thesis. Submits claims to the HoD for honorarium. Updates profile for easy visibility.
- **Head of Department (HoD)** - Assigns Internal Evaluators, evaluates submissions from Supervisors and submits to the Faculty Postgraduate Committee Representative (FPGC-R). Proposes External Evaluators.
- **FPGC-R** - Member of the HDC headed by the DVC TLU. Evaluates, submits and present HoD's submissions (thesis or SoPs), or any other related matters for discussion to the HDC.
- **Faculty Postgraduate Committee (FPGC)** - Selects students to supervise from postgraduate applications. Assigns external examiners from whom the HoD proposes.

NB: All *italicised* words/phrases are names of template files are names files shared via Moodle. Do NOT reproduce them, but let them act as a guide.

Proposed Solution The proposed solution consist of a web and mobile solution with the following modules/teams: Student, Supervisor, External Evaluator, HoD, FPGC-R, FPGC, Backend, Mobile, and DevOps. Take note that this is one big project even though you will be working mostly in smaller teams.

The assessment rubric is summarised in Table 1 and expounded below:



Table 1: Assessment Rubric

Criteria / Component	Poor (9)	Average (55)	Excellent (100)
Requirements Elicitation	(1)	(5)	(10)
User Interaction	(1.5)	(8)	(15)
Architecture	(1.5)	(11)	(20)
Code	(1.5)	(8)	(15)
Documentation	(1)	(6)	(10)
Individual Contribution	(1.5)	(11)	(20)
Presentation	(1)	(6)	(10)

NB: Excellent means all components were met satisfactorily. The scale goes down to Poor.

1. **Requirements Elicitation** - Improving on the requirements given in this document. This requires consuming supporting documents & engaging stakeholders.
2. **User Interaction** - Aesthetics, navigation, ease of use, leveraging platform/language libraries.
3. **Architecture** - Layered architecture, separation of concerns, modularisation.
4. **Code** - Easy to read code, comments, conformance to language conventions, Units tests.
5. **Documentation** - Technical specification.
6. **Individual Contribution** - Github logs.
7. **Presentation** - team presentation, clarity, confidence, Q&A.

This paper consists of 2 pages (including the front page).